

Physiological roles of endocytosis and presynaptic scaffold in vesicle replenishment at fast-signaling and slow-plastic synapses

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Abstract

After exocytosis, release sites are cleared of vesicular residues to be replenished with transmitter-filled vesicles. Endocytic and scaffold proteins are thought to underlie this mechanism. However, physiological significance of the site-clearance mechanism among diverse central synapses remains unknown. Here, we tested this using action-potential evoked EPSCs at fast-signaling calyx and slow-plastic hippocampal CA1 synapses in post-hearing mice brain slices at physiological temperature and Ca^{2+} . Pharmacological block of endocytosis enhanced synaptic depression at the calyx synapse, whereas it attenuated synaptic facilitation at the hippocampal synapse. Block of scaffold protein activity likewise enhanced synaptic depression at the calyx but had no effect at the hippocampal synapse. At the calyx, enhancement of synaptic depression by blocking endocytosis or scaffold activity occurred at nearly identical time courses with a time constant of several milliseconds starting immediately after the stimulation onset. Neither endocytic nor scaffold protein inhibitors prolonged the recovery from short-term depression. We conclude that endocytic release-site clearance can be a universal phenomenon supporting vesicle replenishment across fast-signaling and slow-plastic synapses, whereas presynaptic scaffold mechanism likely plays a specialized role in vesicle replenishment predominantly at fast synapses.

eLife assessment

Following synaptic vesicle fusion events at release sites, vesicle remnants will need to be cleared in order to allow new rounds of vesicle docking and fusion. This **fundamental** study of Mahapatra and Takahashi examines the role of release site clearance in synaptic transmission during repetitive activity in two types of central synapses, the giant calyx of Held and hippocampal CA1 synapses. The study uses pharmacological approaches to interfere with release site clearance by blocking membrane retrieval (endocytosis). They compare the effects on short-term plasticity with those obtained by pharmacologically inhibiting scaffold protein activity. The data presented make a **compelling** case for fast endocytosis as necessary for rapid site clearance and vesicle recruitment to active zones. The data reveal an unexpected, fast role for local site clearance in counteracting synaptic depression.

Introduction

Chemical synaptic transmission depends upon fusion of transmitter-filled vesicles with the presynaptic membrane. At presynaptic terminals, there are a limited number of vesicular release sites, which can become refractory while discharged vesicles are remaining at the site (Katz, 1993 [↗](#)), inhibiting subsequent vesicle fusion. Following inactivation of the temperature-sensitive endocytic protein dynamin in *Drosophila* mutant *Shibire*, short-term depression (STD) of neuromuscular transmission is enhanced within tens of milliseconds of the onset of stimulation (Kawasaki et al., 2000 [↗](#)), suggesting that vesicular endocytosis plays a significant role in release site-clearance, in addition to its well-established role in the vesicle recycling (Neher and Sakaba, 2008 [↗](#)). At the calyx of Held, in slices from pre-hearing rats, pharmacological block of endocytosis slightly enhances STD and markedly prolongs the recovery from STD (Hosoi et al., 2009 [↗](#)). Likewise, at cultured hippocampal synapses, blocking endocytosis enhances depression at high-frequency stimulation due to slow clearance of vesicular component from release-sites (Hua et al., 2013 [↗](#)). Genetic ablation of the endocytic adaptor protein AP-2 μ (Jung et al., 2015 [↗](#)), synaptophysin (Rajappa et al., 2016 [↗](#)), or the secretory carrier membrane protein SCAMP5 (Park et al., 2018 [↗](#)) enhances synaptic depression in cultured cells, in agreement with the site-clearance role of endocytosis.

The presynaptic scaffold protein intersectin is a guanine nucleotide exchange factor, which activates the Rho-GTPase CDC42, thereby regulating the filamentous (F) actin assembly (Hussain et al., 2001 [↗](#); Marie et al., 2004 [↗](#)). At the calyx of Held in slices from pre-hearing rodents, genetic ablation of intersectin 1 or pharmacological block of CDC42 abolishes the fast component of the rate of recovery from STD (Sakaba et al., 2013 [↗](#)), suggesting that the scaffold protein cascade, comprising intersectin, CDC42 and F-actin, contributes to rapid vesicle replenishments possibly through release site-clearance (Japel et al., 2020 [↗](#)).

Although the site-clearance hypothesis is well supported, its physiological significance among diverse central synapses remains unknown. To address this, we evoked EPSCs by afferent fiber stimulations in post-hearing (P13-15) mice brain slices, without intra-terminal perturbation, at brainstem calyces of Held and at hippocampal CA1 synapses at physiological temperature (PT, 37°C; Sanchez-Alavez et al., 2011) and in artificial cerebrospinal fluid (aCSF) containing 1.3 mM- Ca^{2+} , in reference to rodent CSF (Jones and Keep, 1988 [↗](#); Silver and Erecinska, 1990 [↗](#); Inglebert et al., 2020 [↗](#)). The post-hearing calyx of Held is a fast-signaling relay synapse that can respond to inputs up to 500 Hz without failure *in vivo* (Sonntag et al., 2009 [↗](#)) and displays short-term depression during repetitive stimulation. In contrast, the hippocampal CA1 synapse is a relatively slow synapse that starts to fail transmission from Schaffer collateral (SC) to CA1 pyramidal cells (PC) at 40 Hz in slice (Combe et al., 2018 [↗](#)) and exhibits short-term facilitation and long-term plasticity. At these different synapses, we tested the effect of blocking endocytosis or scaffold protein cascade activity on EPSCs evoked by a train of repetitive stimulations (30x) at different frequencies. These experiments indicated that the endocytosis-driven site-clearance mechanism activity-dependently supports vesicle replenishment, thereby counteracting synaptic depression at brainstem calyceal synapses and boosting synaptic facilitation at hippocampal SC-CA1 synapses. In contrast, the vesicle replenishing role of the scaffold mechanism is independent of activity and restricted to fast calyceal synapses, where it co-operates with the endocytosis-dependent site-clearance, thereby maintaining synaptic strength and enabling high-fidelity fast neurotransmission.

Results

Potency of endocytic blockers on slow and fast endocytosis at the calyx of Held

To clarify physiological roles of endocytosis in high-frequency synaptic transmission, we first examined the effect of different endocytic inhibitors on fast and slow endocytosis, using presynaptic membrane capacitance measurements (Sun et al., 2004; Yamashita et al., 2010) at post-hearing calyces of Held (P13-15) at PT (37°C; Renden and von Gersdorff 2007) in aCSF containing 2.0 mM $[Ca^{2+}]$ (Figure 1) to ensure sufficient Ca^{2+} currents to induce exo-endocytosis. Dynasore is a dynamin-dependent endocytosis blocker (Macia et al., 2006; Newton et al., 2006) blocking both clathrin-mediated and clathrin-independent endocytosis (Park et al., 2013; Delvendahl et al., 2016). A second endocytosis blocker Pitstop-2 is thought to preferentially block clathrin-mediated endocytosis (von Kleist et al., 2011; Delvendahl et al., 2016; Lopez-Hernandez et al., 2022) but it reportedly blocks clathrin-independent endocytosis as well (Dutta et al., 2012; Willox et al., 2014). Since genetic ablation of endocytic proteins can be associated with robust compensatory effects (Ferguson et al., 2009; Park et al., 2013), we adopted acute pharmacological block using endocytic inhibitors Dynasore and Pitstop-2.

At calyces of Held, a single short (5 ms) command pulse elicited an exocytic capacitance jump (ΔC_m) followed by a slow endocytic decay rate (29 fF/s, Figure 1A). In the presence of Dynasore (100 μ M) or Pitstop-2 (25 μ M) in the perfusate (within 10-60 min of application) this slow endocytosis showed a strongly prolonged time course without accompanied by the change of ΔC_m or presynaptic Ca^{2+} currents (Table S1). In a repetitive stimulation protocol, the accelerating endocytosis can be induced by a 1-Hz train of 20-ms square pulses (Wu et al., 2005; Yamashita et al., 2010). Dynasore or Pitstop-2 significantly inhibited the maximal rate of 250 fF/s of this accelerated fast endocytosis (20 ms x 15 times at 1 Hz) to ~ 50 % (Figure 1B, Table S1). In another stimulation protocol (Figure 1C), 10 command pulses (20 ms duration) applied at 10 Hz induced a fast endocytosis with a decay rate (350 fF/s) by more than 10 times faster than the slow endocytosis elicited by a 5 ms pulse (Figure 1A). Dynasore or Pitstop-2 significantly inhibited the rate of this fast endocytosis (Figure 1C, Table S1) too. The potencies of bath-applied Dynasore or Pitstop-2 for blocking slow and fast forms of endocytosis were comparable to that of dynamin-1 proline-rich domain peptide (Dyn-1 PRD peptide) directly loaded in calyceal terminals (1mM; Figure S1; Yamashita et al., 2005). Thus, at calyces of Held, bath-application of Dynasore or Pitstop-2 can block both fast and slow endocytosis without perturbing presynaptic intracellular milieu.

Effects of endocytic blockers on synaptic depression and recovery from depression at brainstem calyceal synapses

Using Dynasore or Pitstop-2, we investigated the effect of blocking endocytosis on synaptic transmission at the calyx of Held in brainstem slices from post-hearing mice (P13-15). At PT (37°C) in physiological aCSF (1.3mM $[Ca^{2+}]$), EPSCs were evoked by a train of 30 stimulations applied to afferent fibers at 10 or 100 Hz (Figure 2). Neither Dynasore nor Pitstop-2 (in the perfusate, within 10-60 min of application) affected the first EPSC amplitude in the train (Figure S2A). During a train of stimulations at 10 Hz, in the presence of Dynasore (100 μ M), synaptic depression was slightly enhanced (from 45% to 52%, by 1.15 times), whereas Pitstop-2 had no effect (Figure 2A). At 100-Hz stimulations, however, both Dynasore and Pitstop-2 markedly and equally enhanced synaptic depression within 10 ms from the stimulus onset, increasing the magnitude of steady state STD (averaged from #26-30 EPSCs) from 58% to 75% (1.3 times; $p < 0.001$, t-test; Figure 2B). These results suggest that endocytosis-dependent rapid SV replenishment operates during high-frequency transmission at the mammalian central synapses like at the neuromuscular junction (NMJ) of *Drosophila* (Kawasaki et al., 2000). Since glutamate released during action potential evoked EPSCs does not desensitize or saturate postsynaptic receptors at post-hearing

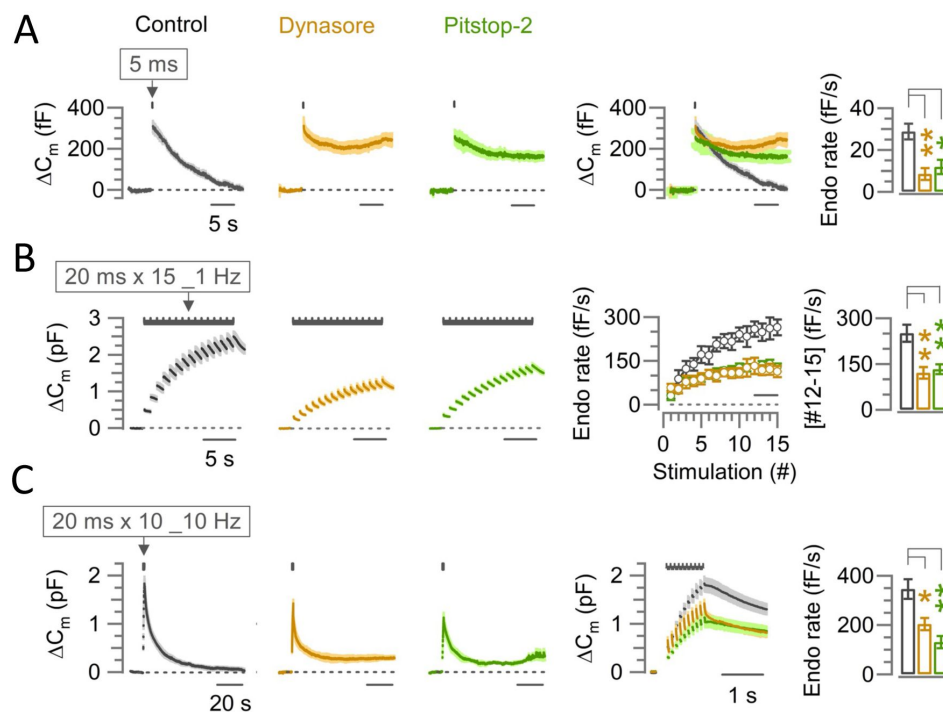


Figure 1.

The endocytic blocker Dynasore or Pitstop-2 inhibits slow, fast-accelerating and fast endocytosis at the calyx of Held

(A) Average traces of slow endocytic membrane capacitance changes (ΔC_m) in response to a 5-ms depolarizing pulse (stepping from -70 mV to +10 mV) in the absence (control, black trace) or presence of Dynasore (100 μ M, 10–60 min, brown trace) or Pitstop-2 (25 μ M, 10–60 min, green trace), recorded from the calyx of Held presynaptic terminal in slices from P13–15 mice at physiological temperature (PT, 37°C) and in 2.0 mM Ca^{2+} aCSF. The 4th panel from the left shows the superimposed average ΔC_m traces under control, Dynasore and Pitstop-2. The rightmost bar graph shows the endocytic decay rate (calculated from the slope 0.45–5.45 s after stimulation) that was slower in the presence of Dynasore (8.5 ± 2.8 fF/s; $n = 5$; $p = 0.004$, Student's t-test) or Pitstop-2 (11.9 ± 3.4 fF/s; $n = 5$; $p = 0.015$) than control (28.8 ± 3.7 fF/s; $n = 5$). Significance level was set at $p < 0.05$, denoted with asterisks (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

(B) Fast-accelerating endocytosis induced by a train of 20-ms depolarizing pulses (repeated 15 times at 1 Hz) in the absence (control) or presence of Dynasore or Pitstop-2. Averaged traces are shown as in (A). The 4th panel shows the endocytic rates (fF/s) calculated from the slope of C_m decay, 0.45–0.95 ms after each stimulation pulse under control, Dynasore, or Pitstop-2 (superimposed). The rightmost bar graph shows the endocytic rate averaged from stimulations #12–15 (bar in 4th panel) that was slower in the presence of Dynasore (121 ± 19 fF/s; $n = 6$; $p = 0.0045$, Student's t-test) or Pitstop-2 (133 ± 16 fF/s; $n = 5$; $p = 0.007$) than control (251 ± 28 fF/s; $n = 6$), indicating significant inhibition of the fast-accelerating endocytosis by Dynasore or Pitstop-2 (also see Table S1).

(C) Fast-endocytosis (average traces) evoked by a train of 20-ms pulses (repeated 10 times at 10 Hz) in the absence (control) or presence of Dynasore or Pitstop-2. The 4th panel shows cumulative ΔC_m traces (superimposed) in an expanded timescale during and immediately after the 10-Hz train. The rightmost bar graph indicates endocytic decay rates (measured 0.45–1.45 s after the 10th stimulation) in Dynasore (204 ± 25.3 fF/s; $n = 4$; $p = 0.02$; Student's t-test) or in Pitstop-2 (131 ± 24.6 fF/s; $n = 4$; $p = 0.002$) both of which were significantly slower than control (346 ± 40.1 fF/s; $n = 6$).

calyces of Held (Ishikawa et al, 2002 [↗](#)) unlike at pre-hearing calyces (Yamashita et al, 2009 [↗](#)), enhanced synaptic depression in the presence of endocytic blockers during repetitive fiber stimulations is most likely caused by pre-synaptic mechanisms. In fact, the results of endocytic block were essentially the same in the absence (**Figure 2** [↗](#)) or presence (**Figure 3S** [↗](#)) of the low-affinity glutamate receptor ligand kynurenic acid (1 mM), which reduces postsynaptic receptor occupancy with glutamate.

At pre-hearing calyces of Held at room temperature (RT), in 2.0 mM $[Ca^{2+}]$ solution, with a long command pulse stimulation under voltage-clamp, dynamin inhibitors markedly prolong the recovery of EPSCs from STD (Hosoi et al., 2009 [↗](#)). In contrast, at post-hearing calyces in 1.3 mM $[Ca^{2+}]$ and at PT (37°C), in the presence of the endocytic blockers, recovery of EPSCs from STD showed no prolongation (**Figure 2A** [↗](#), 2, 2B2, Table S2). Conversely, the recovery from STD caused by 100-Hz stimulation was significantly accelerated at both fast and slow recovery time constants (**Figure 2B** [↗](#), Table S2). To explore the reason for these different results, we raised Ca^{2+} concentration in aCSF to 2.0 mM at PT (Figure S4A) or RT (Figure S4B) or raised stimulation frequency to 200 Hz at PT (Figure S4C). None of these manipulations caused significant change in the recovery from STD (Figure S4). Remaining difference may include the species and age differences between animals (pre-hearing rats vs post-hearing mice) and stimulation intensity that is stronger in the whole-cell voltage-clamp command pulse protocol than afferent fiber stimulation. Lack of recovery acceleration in 2.0 mM $[Ca^{2+}]$ (Figure S4) by Dynasore or Pitstop-2 is consistent with the observation in dynamin-1 knockout mice (Mahapatra et al, 2016 [↗](#)) suggesting that recovery acceleration of unknown mechanism observed in physiological $[Ca^{2+}]$ and temperature (**Figure 2B** [↗](#)) was likely masked in 2.0 mM $[Ca^{2+}]$.

Effects of scaffold protein cascade inhibitors on vesicle endocytosis, synaptic depression, and recovery from depression at brainstem calyceal synapses

Among presynaptic scaffold proteins, F-actin is assembled by the Rho-GTPase CDC42 activated by the scaffold protein intersectin, a guanine nucleotide exchange factor (Hussain et al., 2001 [↗](#); Marie et al., 2004 [↗](#)). Genetic ablation of intersectin 1 or pharmacological inhibition of CDC42 activity prolongs the recovery from STD at pre-hearing calyces of Held, suggesting that this scaffold cascade contributes to vesicle replenishment, possibly via release site-clearance (Sakaba et al., 2013 [↗](#); Japel et al., 2020 [↗](#)).

Genetic ablation of intersectin inhibits endocytosis in cultured synapses (Yu et al., 2008 [↗](#)), but does not affect endocytosis at pre-hearing calyces of Held (Sakaba et al., 2013 [↗](#)). We examined the effect of blocking CDC42 using ML 141 (10 μ M, 10-60 min) or inhibiting F-actin assembly using Latrunculin-B (15 μ M, 10-60 min) in perfusates on fast and slow endocytosis at the post-hearing (P13-15) calyx of Held at PT (**Figure 3** [↗](#)) in aCSF containing 2.0 mM $[Ca^{2+}]$. In agreement with previous report (Sakaba et al., 2013 [↗](#)), none of these inhibitors affected the fast or slow forms of endocytosis.

We then recorded EPSCs in 1.3 mM $[Ca^{2+}]$ at PT. Neither ML141 nor Latrunculin-B affected the first EPSC amplitude in the train (Figure S2A), alike Latrunculin-B effects at RT and 2.0 mM $[Ca^{2+}]$ (Mahapatra et al., 2016 [↗](#); Mahapatra and Lou, 2017 [↗](#)). During a train of stimulations at 10 Hz (**Figure 4A** [↗](#) 1) or 100 Hz (**Figure 4B** [↗](#) 1), in the presence of ML141 or Latrunculin-B in perfusates, synaptic depression was markedly and equally enhanced, with STD magnitude (#26-30) increasing from 45% to 62% (1.4 times; $p < 0.001$, t-test; **Figure 4A** [↗](#) 1) at 10 Hz and from 58% to 80% at 100 Hz (1.4 times; $p < 0.001$, t-test; **Figure 4B** [↗](#) 1). Thus, in contrast to endocytic blockers (**Figure 2** [↗](#)), the depression-enhancing effects of scaffold cascade inhibitors were not activity-dependent.

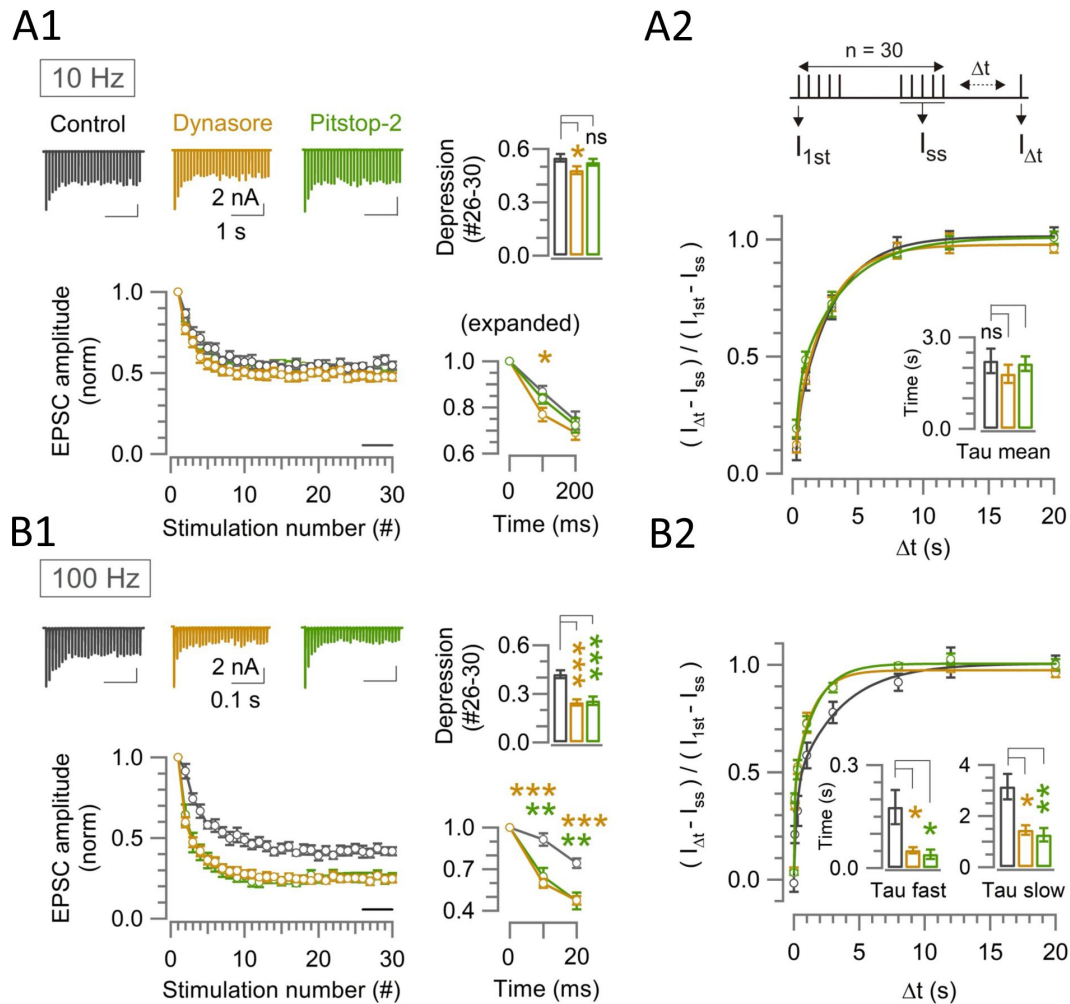


Figure 2.

Endocytic blockers rapidly enhance synaptic depression in a stimulation frequency-dependent manner, but do not prolong the recovery from depression at the calyx.

(A1, B1) A train of 30 EPSCs were evoked at the calyx of Held by afferent fiber stimulation at 10 Hz (A1) or 100 Hz (B1) in the absence (control, black traces) or presence of Dynasore (10-60 min, brown traces) or Pitstop-2 (10-60 min, green traces) at PT (37°C) in 1.3 mM Ca^{2+} aCSF. *Panels from left to right:* left-top: sample EPSC traces; left-bottom: normalized average EPSC amplitudes at each stimulus number; right-top bar graph: steady-state depression of EPSC amplitudes (mean of EPSCs #26-30, bar in the second panel); right-bottom: 1st-3rd EPSC amplitudes in expanded timescale.

(A2, B2) The recovery of EPSCs from STD in control, or in the presence of Dynasore or Pitstop-2 at the calyx of Held measured using a stimulation protocol (shown on top in A2); a train of 30 stimulations at 10 Hz (A2) or 100 Hz (B2) followed by test pulses after different time intervals (Δt : 0.02, 0.1, 0.3, 1, 3, 8, 12, and 20s). The EPSC amplitude after Δt ($I_{\Delta t}$) relative to the first EPSC in the stimulus train (I_{1st}) was normalized by subtracting the steady state EPSCs (I_{ss}) to measure the recovery rates.

(A1) During 10-Hz stimulation, the steady-state depression under control (0.55 ± 0.02 ; $n = 11$) was slightly enhanced in the presence of Dynasore (0.48 ± 0.02 ; $n = 12$; $p = 0.03$, Student's t-test) but not in Pitstop-2 (0.53 ± 0.02 ; $n = 7$; $p = 0.4$, no significant difference).

(A2) After 10-Hz stimulation, the time constant of EPSCs recovery in control (2.3 ± 0.4 s; $n = 9$) was unchanged in the presence of either Dynasore (1.7 ± 0.2 s; $n = 8$; $p = 0.2$, Student's t-test) or Pitstop-2 (1.9 ± 0.3 s; $n = 7$; $p = 0.4$).

(B1) During 100-Hz stimulation, EPSCs underwent a significant depression starting at the 2nd stimulation (10 ms) in Dynasore (0.6 ± 0.03 ; $n = 10$; $p < 0.001$, t-test) or Pitstop-2 (0.65 ± 0.06 ; $n = 7$; $p = 0.005$) than control (0.91 ± 0.05 ; $n = 11$). Bar graph indicates STD magnitudes; control: (0.42 ± 0.025), Dynasore: (0.25 ± 0.02 ; $p < 0.001$), and Pitstop-2: (0.26 ± 0.03 ; $p < 0.001$).

(B2) After 100-Hz stimulation, both fast and slow recovery time constants were significantly faster in the presence of Dynasore or Pitstop-2 than control (bar graphs, Table S2).

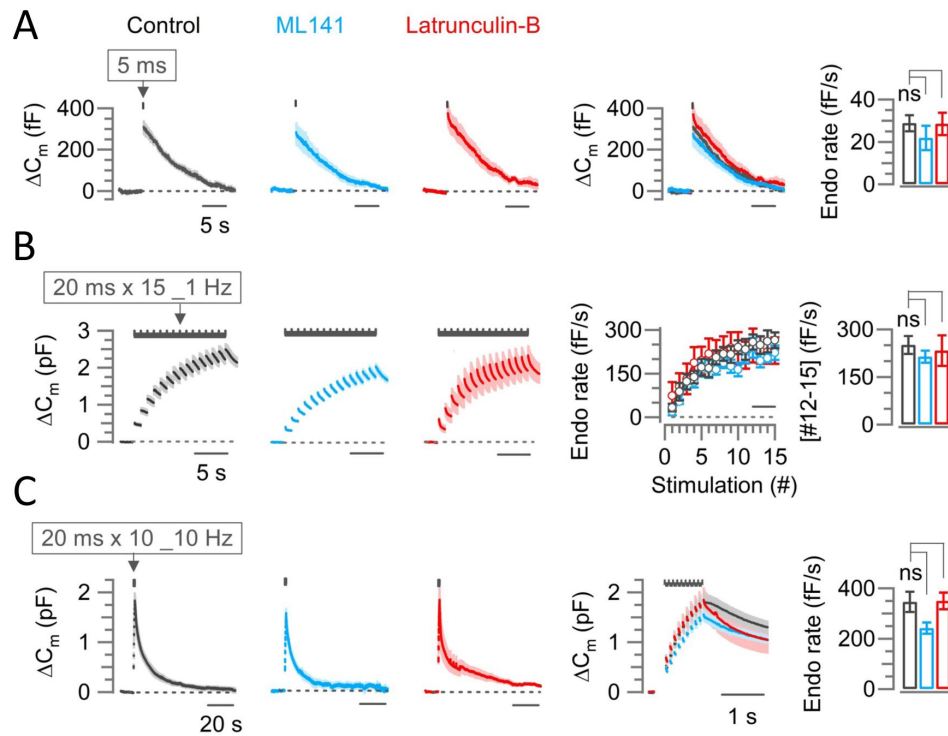


Figure 3.

Scaffold machinery inhibitors have no effect on endocytic membrane retrievals at the calyx of Held

The CDC42 inhibitor ML141 (10 μ M, 10–60 min, cyan) or actin depolymerizer Latrunculin B (Lat-B, 10–60 min, 15 μ M, red) had no effect on slow endocytosis in response to a 5-ms depolarizing pulse (A) or on fast accelerating endocytosis (B; induced by 1-Hz train of 20 ms x 15 pulses) or fast endocytosis (C; evoked by a 10-Hz train of 20 ms x 10 pulses) at PT and in 2.0 mM Ca^{2+} aCSF.

(A) Averaged and superimposed ΔC_m traces in response to a 5-ms pulse. The rightmost bar graph indicates the endocytic decay rate in control (28.8 ± 3.7 fF/s; $n = 5$), unchanged by ML141 (21.9 ± 5.8 fF/s; $n = 6$; $p = 0.42$, Student's t-test) or Lat-B (28.5 ± 5.2 fF/s; $n = 6$; $p = 0.97$, Table S1).

(B) The average endocytic rate (#12–15) in control (251 ± 28 fF/s; $n = 6$) unaltered by ML141 (214 ± 19 fF/s; $n = 5$; $p = 0.3$, Student's t-test) or Latrunculin-B (233 ± 48 fF/s; $n = 4$; $p = 0.7$; Table S1).

(C) Fast endocytic decay rate in the presence of ML141 (242 ± 21.5 fF/s; $n = 4$; $p = 0.054$, t-test) or Lat-B (350 ± 33.3 fF/s; $n = 5$; $p = 0.95$) was not different from control (346 ± 40.1 fF/s; $n = 6$; Table S1).

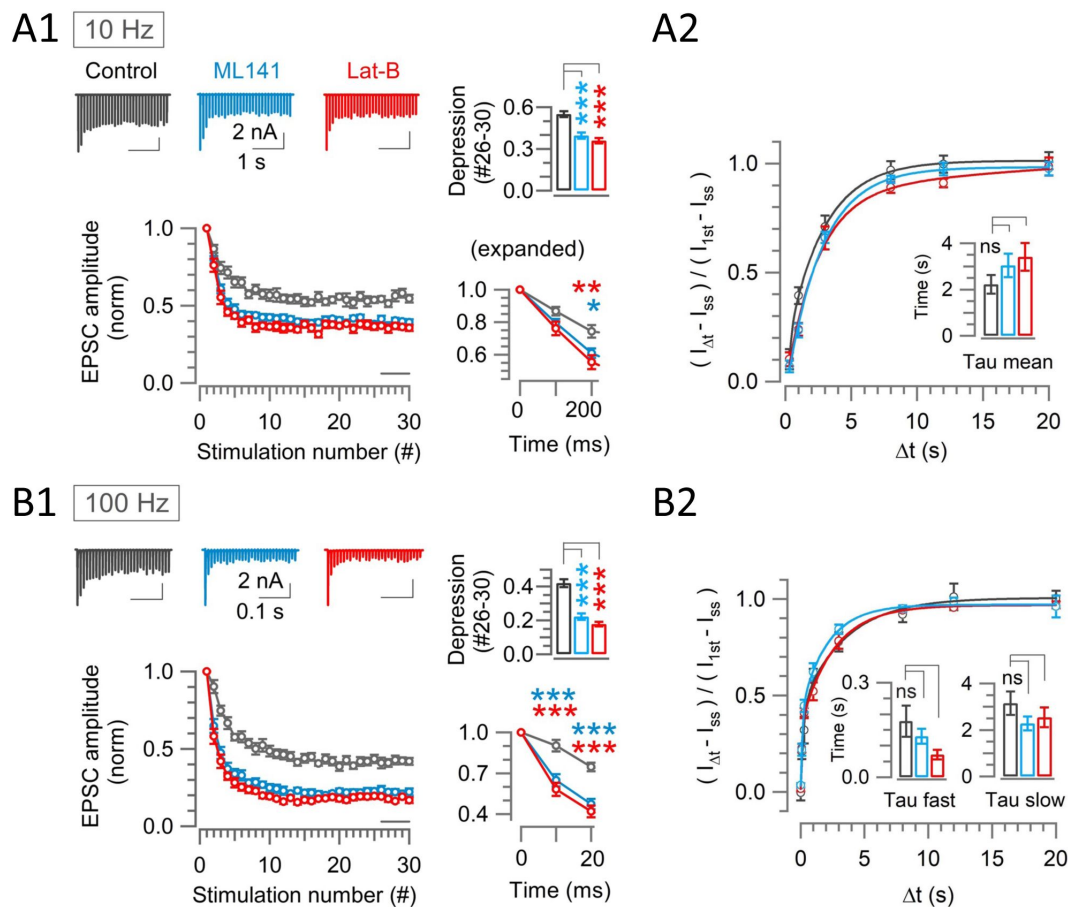


Figure 4.

Scaffold machinery inhibitors strongly enhance synaptic depression in a stimulation frequency-independent manner, without prolonging the recovery from depression at the calyx

(A1, B1) EPSCs (30x) evoked at the calyx of Held by afferent fiber stimulation at 10 Hz (A1) or 100 Hz (B1) in the absence (control, black) or presence of ML141 (10-60 min, cyan) or Lat-B (10-60 min, red) at PT and in 1.3 mM Ca^{2+} aCSF. *Panels from left to right: like Figure 2*.

(A2, B2) The recovery of EPSCs from STD in control, or in the presence of ML141 or Latrunculin-B at the calyx of Held measured at 10 Hz (A2) or 100 Hz (B2) – like *figure 2*.

(A1) At 10-Hz stimulation, enhancement of depression became significant from the 3rd stimulation (200 ms) in the presence of ML141 (0.61 ± 0.03 ; $n = 9$; $p = 0.013$, Student's t-test) or Lat-B (0.55 ± 0.04 ; $n = 7$; $p = 0.006$) compared to control (0.74 ± 0.04 , $n = 11$). Bar graph indicates the steady-state depression (STD) strongly enhanced from control (0.55 ± 0.02) by ML141 (0.4 ± 0.02 ; $p < 0.001$) or Lat-B: (0.36 ± 0.02 ; $p < 0.001$).

(A2) After a train of 30 stimulations at 10 Hz, the time constant of EPSCs recovery under control was unchanged by ML141 or Latrunculin-B (Table S2).

(B1) At 100-Hz stimulation, EPSCs showed significant enhancement of depression at the 2nd stimulation (10 ms) in the presence of ML141 (0.65 ± 0.05 ; $n = 10$; $p < 0.001$) or Lat-B (0.58 ± 0.05 ; $n = 8$; $p < 0.001$) from control (0.91 ± 0.05 ; $n = 11$). Bar graph indicates strong STD produced by ML141 (0.22 ± 0.02 ; $p < 0.001$) or Lat-B (0.18 ± 0.013 ; $p < 0.001$) compared to control (0.42 ± 0.025).

(B2) The time course of EPSC recovery from STD induced by a train of 30 stimulations at 100 Hz, indicating no significant change caused by ML141 or Lat-B in the recovery time constants (Table S2). The stimulation and recovery protocols were the same as those used in *Figure 2*.

Like endocytic blockers, scaffold cascade inhibitors did not prolong the recovery from STD at 10 Hz (**Figure 4A** [2](#)) or at 100 Hz (**Figure 4B** [2](#)). These results contrast with those previously reported at pre-hearing calyces (Sakaba et al, 2013), where genetic ablation of intersectin or pharmacological block of CDC42 prolongs the recovery from STD induced by double command pulse stimulation under voltage-clamp.

Kinetic comparison of synaptic depression enhanced by endocytic blockers and scaffold cascade inhibitors and evaluation of their combined effect

During 100-Hz stimulation, enhancement of synaptic depression by endocytic blockers or scaffold cascade inhibitors are prominent already in the second EPSCs (**Figure 2B** [1](#), [4B1](#)). Exponential curve fits to data points (**Figure 5A** [2](#)) indicated that the EPSCs underwent a single exponential depression in controls with a mean time constant of 37 ms ($n = 11$), whereas in the presence of endocytic blockers or scaffold cascade inhibitors, EPSCs amplitude underwent double exponential decay with a fast time constant of 5–10 ms and a slow time constant of tens of milliseconds, the latter of which was similar to the time constant in control (**Figure 5B** [2](#)). These data indicate that endocytic and scaffold mechanisms for vesicle replenishment operate predominantly at the beginning of synaptic depression. This is consistent with a lack of prolongation in the recovery from STD by endocytic blockers or scaffold cascade inhibitors (**Figure 2A** [2](#), [2B2](#); **Figure 4A** [2](#), [4B2](#)).

Since the enhancement of synaptic depression by endocytic blockers or scaffold cascade inhibitors occurred mostly at the early phase of synaptic depression, endocytosis- and scaffold-dependent synaptic strengthening likely operate simultaneously during high-frequency transmission. To clarify whether these mechanisms are additive or complementary, we co-applied Dynasore and Latrunculin-B during 100 Hz stimulation (**Figure 5C** [2](#)). The magnitude of EPSC depression by the co-application was the same as that by single Latrunculin-B application (**Figure 5C** [2](#)), suggesting that endocytosis and scaffold mechanisms complementarily maintain synaptic strength during high-frequency transmission.

The F-actin depolymerizers Latrunculin A and Latrunculin B are widely utilized to examine presynaptic roles of F-actin. However, the results of STD and recovery can vary according to the application methods and experimental conditions (**Figure S5A**). At the calyx of Held, in 2.0 mM $[Ca^{2+}]$ aCSF at PT, Latrunculin A perfusion (20 μ M, 1h, **Figure 5A** [1](#)) showed a significant enhancement of STD at 100 Hz within 10–60 min of drug application (**Figure S5B1**, **S5D**), as previously reported for Latrunculin B perfusion at RT and 200 Hz (Mahapatra et al, 2016 [2](#)). However, 1h preincubation of slices with Latrunculin A (20 μ M) followed by perfusion with standard aCSF (~1h, **Figure S5A2**) had no significant effect on STD induced by a train of 30 stimuli at 100 Hz (**Figure S5B2**, **S5D**), as previously reported (Babu et al, 2020 [2](#)). In 1.3 mM $[Ca^{2+}]$ aCSF at PT (**Figure 5C** [2](#)), marked STD enhancement was observed by continuous perfusion of Latrunculin B (15 μ M, **Figure S5C**, **Figure S5D**) or Latrunculin A (20 μ M, **Figure S5C**, **Figure S5D**). Thus, Latrunculin A (20 μ M) and Latrunculin B (15 μ M) are equipotent for STD enhancement in these experiments (**Figure S5C**, **S5D**).

When the data are compared between 1.3 mM $[Ca^{2+}]$ and 2.0 mM $[Ca^{2+}]$, STD levels after Latrunculin A perfusion are not different ($p > 0.05$, **Figure S5D**), whereas control STD level in 1.3 mM $[Ca^{2+}]$ (~0.4) was significantly less than 2.0 mM $[Ca^{2+}]$ (~0.3; $p = 0.0036$, t-test; **Figure S5D**). Thus, physiological 1.3 mM $[Ca^{2+}]$ revealed a robust STD-counteracting function of F-actin.

Regarding the recovery from STD, Ca^{2+} -dependent recovery (CDR) component of tens of ms time constant is produced by strong stimulation at the calyx of Held at RT (Wang and Kaczmarek, 1998 [2](#)) and inhibited by Latrunculin B (Lee et al, 2013). CDR is also observed after STD induced by 100 Hz stimulation at PT and blocked by Latrunculin A pretreatment (20 μ M) in 2 mM $[Ca^{2+}]$ (Babu

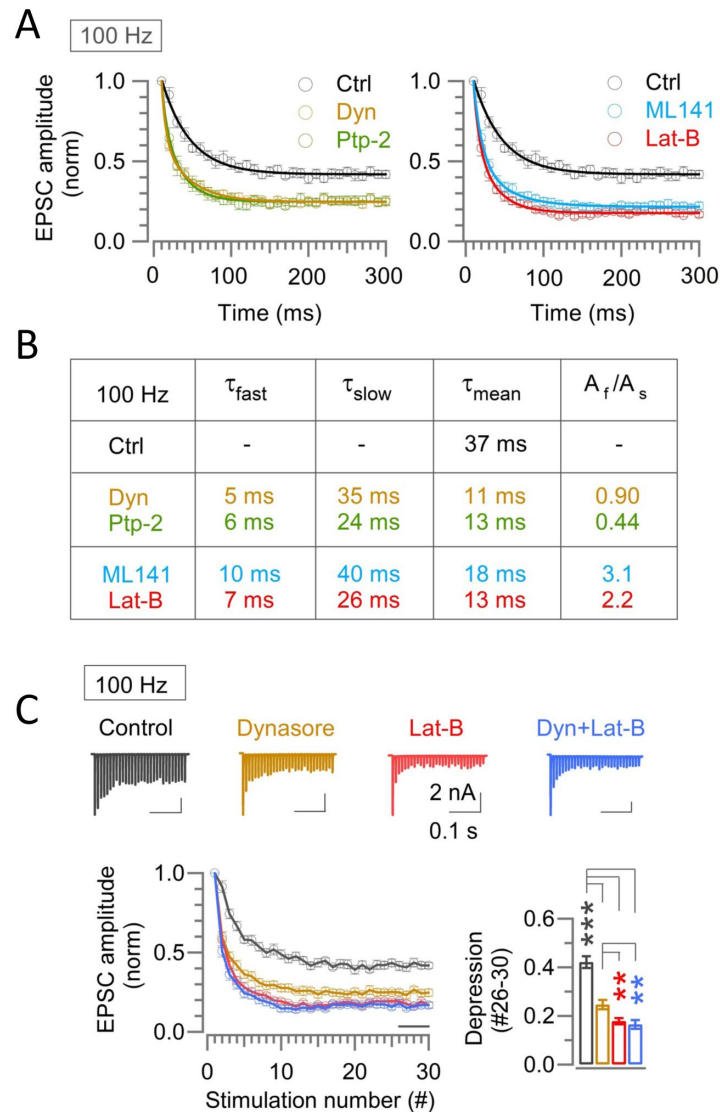


Figure 5.

Endocytic and scaffold machineries co-operate for rapid vesicle replenishment during high-frequency transmission at the calyx of Held.

(A) Exponential curve fits to the time-course of synaptic depression during 100-Hz stimulation under control and in the presence of either endocytic blockers or scaffold cascade inhibitors. The control time-course was best fit to a single exponential, whereas the time-course in the presence of endocytic blockers or scaffold cascade blockers was fit best to double exponential function with fast and slow time constants.

(B) Parameters for the curve-fit, including fast and slow time constants (τ_{fast} , τ_{slow}), weighted mean time constant (τ_{mean}) and relative ratio of fast and slow components (A_f/A_s). Similar fast time constant (τ_{fast}) in the presence of endocytic inhibitors or scaffold cascade inhibitors suggests simultaneous operation of endocytic and scaffold mechanisms for countering the synaptic depression. Similar slow decay time constant irrespective of the presence or absence of blockers suggests that the endocytosis and scaffold mechanisms for vesicle replenishment operates predominantly at the beginning of the high frequency stimulations.

(C) Enhancement of synaptic depression by co-application of Dynasore and Latrunculin-B (10-60 min) was like Latrunculin-B alone and stronger than Dynasore alone. Sample EPSC traces and normalized depression time courses are shown on the upper and lower left panels, respectively. The STD magnitudes are compared in bar graph; control: (0.42 ± 0.025 ; $n = 11$), Dynasore: (0.25 ± 0.02 ; $n = 10$, $p = 0.009$), Latrunculin-B: (0.18 ± 0.013 ; $n = 8$, $p = 0.005$ vs Dynasore), and Dyn + Lat-B together: (0.17 ± 0.02 ; $n = 11$; $p = 0.005$ vs Dynasore).

et al., 2020 [\[3\]](#)). However, such Latrunculin-sensitive fast recovery component was absent in 1.3 mM $[Ca^{2+}]$ (**Figure 4B** [\[3\]](#), Table S2). Thus, CDR may operate only when excessive Ca^{2+} enters during massive presynaptic activation.

Possible role of endocytosis and scaffold machineries at hippocampal synapses

Counteraction of synaptic depression by endocytosis or scaffold mechanism is highlighted at fast synapses of large structure such as neuromuscular junction or the calyx of Held. However, it remains open whether these mechanisms are conserved at more conventional bouton-type synapses involved in synaptic plasticity. Hence, we recorded EPSCs from hippocampal SC-CA1 pyramidal neurons evoked at high frequency (10 Hz or 25 Hz) by stimulating Schaffer collaterals (SCs) at PT and in aCSF containing 1.3 mM $[Ca^{2+}]$ (**Figure 6** [\[3\]](#)). In this physiologically optimized experimental condition, EPSCs underwent a prominent facilitation (**Figure 6** [\[3\]](#)). In the presence of Dynasore (100 μ M, within 10-60 min of application in perfusate), synaptic facilitation was much less than control, which was noticeable within 300 ms (4th stimulation) at 10 Hz stimulation or within 80 ms (3rd stimulation) at 25 Hz stimulation from the onset (**Figure 6A** [\[3\]](#), 6B). Unexpectedly, Dynasore significantly enhanced the basal EPSC amplitude (Figure S2B), whereas Pitstop-2 had no such effect (Figure S2B). Although less potent than Dynasore, Pitstop-2 significantly attenuated synaptic facilitation (**Figure 6A** [\[3\]](#), 6C). Thus, at hippocampal CA1 synapses, like at fast synapses, endocytosis strengthens synaptic efficacy during high-frequency transmission, thereby boosting short-term synaptic potentiation.

Strikingly, unlike endocytic blockers, neither ML141 nor Latrunculin-B affected the hippocampal short-term facilitation during stimulations at 10 Hz or 25 Hz (**Figure 6C** [\[3\]](#), 6D). Thus, the presynaptic scaffold cascade proteins F-actin or CDC42 unlikely plays a regulatory role on short-term synaptic plasticity at hippocampal synapses. At the CA1 synapse, block of vesicle acidification is reported to enhance synaptic depression at 10-30-Hz stimulation (Ertunc et al., 2007 [\[3\]](#)), presumably because of a lack of filled vesicles undergoing rapid vesicle recycling. However, at hippocampal synapses in culture, vesicle acidification block has no effect on the depression of repetitive exocytosis (Hua et al., 2013 [\[3\]](#)). To determine whether this enhanced depression in the presence of v-ATPase blockers in brain slices (Ertunc et al., 2007 [\[3\]](#)) have any physiological relevance in regulating the synaptic strength during repetitive transmission, we tested the effect of v-ATPase blockers on hippocampal EPSCs. Unlike in 2.0 mM $[Ca^{2+}]$ aCSF at RT (Ertunc et al., 2007 [\[3\]](#)), in 1.3-mM $[Ca^{2+}]$ aCSF at PT, EPSCs showed a prominent facilitation (**Figure 6** [\[3\]](#), Figure S6), and blocking vesicle acidification with Folimycin (67 nM) or Bafilomycin A1 (5 μ M) reduced the basal EPSC amplitudes but had no significant effect on the short-term facilitation (Figure S6). Thus, in physiological conditions, vesicle acidification has no direct effect on the short-term synaptic plasticity at hippocampal CA1 synapses.

Discussion

In acute slices optimized to physiological conditions, blocking endocytosis by bath-application of endocytic blockers enhanced activity-dependent depression of EPSCs evoked by high-frequency stimulation of afferent fibers at brainstem calyceal synapses. The enhancement of depression occurred immediately after the onset of stimulation and proceeded with a several millisecond time constant and then slowly merged into the control time course. Thus, the depression-counteracting effect of endocytosis was restricted to the early epoch of high-frequency transmission when many vesicles occupy the release sites. As number of vesicles undergoing exocytosis during repetitive stimulation gradually reduces, the site-clearance becomes less demanding.

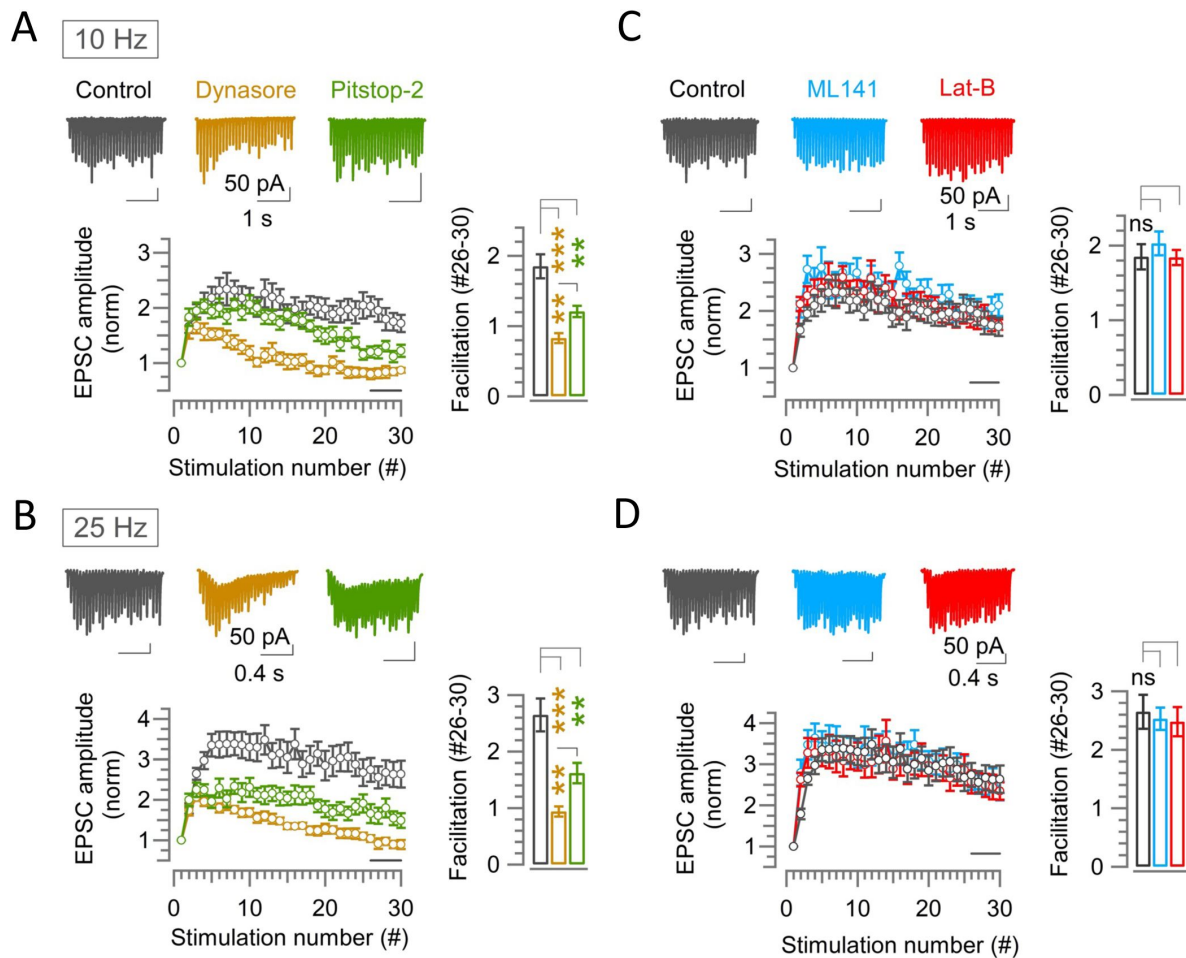


Figure 6.

Endocytic blockers attenuate synaptic facilitation, but scaffold cascade inhibitors have no effect at hippocampal CA1 synapses

(A-D) A train of 30 EPSCs evoked in hippocampal CA1 pyramidal cells by Schaffer collateral stimulation at 10 Hz (A and C) or 25 Hz (B and D) in the absence (control, black) or presence of endocytic blocker Dynasore (100 μ M, 10-60 min, brown) or Pitstop-2 (25 μ M, 10-60 min, green) (A, B), or scaffold protein inhibitor ML141 (10 μ M, 10-60 min, cyan) or Latrunculin-B (15 μ M, 10-60 min, red) (C, D) at PT (37°C) and in 1.3 mM Ca^{2+} aCSF. Top panels show sample EPSC traces. Lower panels show average EPSC amplitudes normalized and plotted against stimulation numbers. Bar graphs show EPSCs amplitudes averaged from #26-30 events.

(A) At 10 Hz stimulation, EPSCs in control showed facilitation reaching a peak at the 7th stimulation (2.34 ± 0.3 ; $n = 17$). At this point, in the presence of Dynasore (1.44 ± 0.15 ; $n = 10$; $p = 0.012$, Student's *t*-test) facilitation was significantly attenuated. The effect of Pitstop-2 (1.85 ± 0.14 ; $n = 14$; $p = 0.06$) was not different. Towards the end of stimulus train (#26-30), synaptic facilitation in control (1.85 ± 0.17) was significantly attenuated by Dynasore (0.83 ± 0.07 ; $p < 0.001$) or Pitstop-2 (1.21 ± 0.08 ; $p = 0.002$).

(B) At 100 Hz stimulation, synaptic facilitation peaked at the 12th stimulation in control (3.5 ± 0.4 ; $n = 16$), at which the facilitation was significantly attenuated by Dynasore (1.6 ± 0.11 ; $n = 11$; $p < 0.001$, *t*-test) or Pitstop-2 (2.11 ± 0.22 ; $n = 14$; $p = 0.004$). Also, at #26-30, synaptic facilitation in control (2.65 ± 0.3) was strongly attenuated by Dynasore (0.94 ± 0.1 ; $p < 0.001$) or Pitstop-2 (1.62 ± 0.2 ; $p = 0.006$).

(C) At 10 Hz stimulation, the peak facilitation at the 7th stimulation in control (2.34 ± 0.3 ; $n = 17$) was not significantly changed by ML141 ($2.5 \pm .22$, $n = 10$; $n = 0.75$, Student's *t*-test) or Lat-B (2.3 ± 0.23 ; $n = 12$; $p = 0.93$). Likewise, the facilitation at #26-30 in control (1.85 ± 0.17) was not altered by ML141 (2.03 ± 0.16 ; $p = 0.44$) or Lat-B (1.84 ± 0.1 ; $p = 0.96$).

(D) At 100 Hz stimulation, peak facilitation at 12th stimulation in control (3.5 ± 0.4 ; $n = 16$) was unchanged by ML141 (3.3 ± 0.34 ; $n = 10$; $p = 0.72$, *t*-test) or Lat-B (3.1 ± 0.4 ; $n = 11$; $p = 0.42$). Facilitation at #26-30 events in control (2.65 ± 0.3) was also unaltered by ML141 (2.53 ± 0.2 ; $p = 0.74$) or Lat-B (2.5 ± 0.25 ; $p = 0.7$).

At the calyx of Held, Dynasore or Pitstop-2 blocked both fast and slow endocytosis (**Figure 1** [↗](#)). The fast endocytosis is characterized with a rate of ~ 350 fF/s, which was slowed by endocytic blockers to ~ 150 fF/s (**Figure 1C** [↗](#), Table S1). This reduction of endocytic rate (200 fF/s) corresponds to 2.54 vesicles/ms (1 fF = 12.7 vesicles; Yamashita et al., 2010 [↗](#); 200 fF/s $\times$ 12.7 vesicles $\times$ 1/1000 ms). Since the mean amplitude of miniature EPSCs, representing a single vesicle response, was ~ 55 pA in 1.3 mM $[Ca^{2+}]$ at PT (data not shown), endocytic block is estimated to reduce EPSC amplitude by ~ 1.4 nA in 10 ms (2.54 vesicles/ms $\times$ 55 pA $\times$ 10 ms $\times$ 1/1000 nA). This is close to the EPSC size difference (1.6 nA) after 10 ms at 100-Hz stimulus onset between the absence and presence of endocytic blockers. Therefore, during high-frequency transmission, the site-clearance by fast endocytosis can fully compensate the initial strong depression. However, the rate of slow endocytosis is 10 times slower than fast endocytosis (**Figure 1A** [↗](#), Table S1), therefore slow endocytosis unlikely contributes to the millisecond-order site-clearance. However, without slow endocytosis, vesicular membrane remaining after spontaneous exocytosis would accumulate and congest release sites. Thus, slow endocytosis may play a house-keeping role in release site-clearance.

At the calyx of Held, scaffold protein inhibitors significantly enhanced the early phase of synaptic depression like endocytic blockers (**Figure 5A** [↗](#), 5B). Unlike endocytic blockers, scaffold protein inhibitors had no effect on endocytosis (**Figure 3** [↗](#)). In contrast to the effect of endocytic blockers (**Figure 2** [↗](#)), the effect of scaffold protein inhibitors was independent of stimulation frequency (**Figure 4** [↗](#)). Co-application of endocytic and scaffold protein cascade inhibitors revealed that the synaptic strengthening effects of endocytosis and scaffold machinery are complementary rather than additive (**Figure 5C** [↗](#)). These results together suggest that fast endocytosis quickly clears vesicle residues, whereas scaffold machinery simultaneously replenishes transmitter-filled vesicles to release sites during high frequency transmission (**Figure 7** [↗](#)). Through this combinatory mechanism synaptic strength and high-fidelity neurotransmission (up to 500 Hz; Sonntag et al., 2009 [↗](#)) can be maintained at fast synapses such as the calyx of Held auditory relay synapse.

At hippocampal CA1 synapses the roles of endocytosis and scaffold machineries in fast vesicle replenishment were different from those at the calyx of Held. In slices optimized to physiological condition, short-term plasticity was depressive at calyceal synapses, whereas facilitatory at hippocampal CA1 synapses. Endocytic blockers attenuated synaptic facilitation at this slow-plastic synapse, suggesting that endocytosis is normally boosting synaptic facilitation (**Figure 6** [↗](#)). Since synaptic strength at facilitatory synapses is determined by a sum of facilitation and depression mechanisms, attenuation of synaptic depression by endocytosis can boost synaptic facilitation. At hippocampal excitatory synapses, short-term facilitation of glutamate release induces long-term potentiation (LTP) of excitatory transmission. Thus, from physiological viewpoints, endocytosis-dependent site clearance can play an essential role in the induction of LTP underlying memory formation.

Strikingly, the scaffold protein inhibitor ML141 or Latrunculin B had no effect on EPSCs at hippocampal CA1 synapses (**Figure 6C** [↗](#), 6D). This is surprising since Latrunculin-A blocks ultra-fast endocytosis in hippocampal culture (Watanabe et al., 2013 [↗](#)). Since Latrunculin-B attenuates vesicle docking at parallel fiber-interneuron synapses in cerebellar slices (Miki et al., 2016 [↗](#)), presynaptic scaffold machinery likely operates at the cerebellar synapses. The scaffold machinery might play vesicle replenishing role at fast-relay synapses, but not at slow-plastic synapses. In contrast, the site-clearance role of endocytosis seems universal among synapses, irrespective of their transmission speed and plasticity.

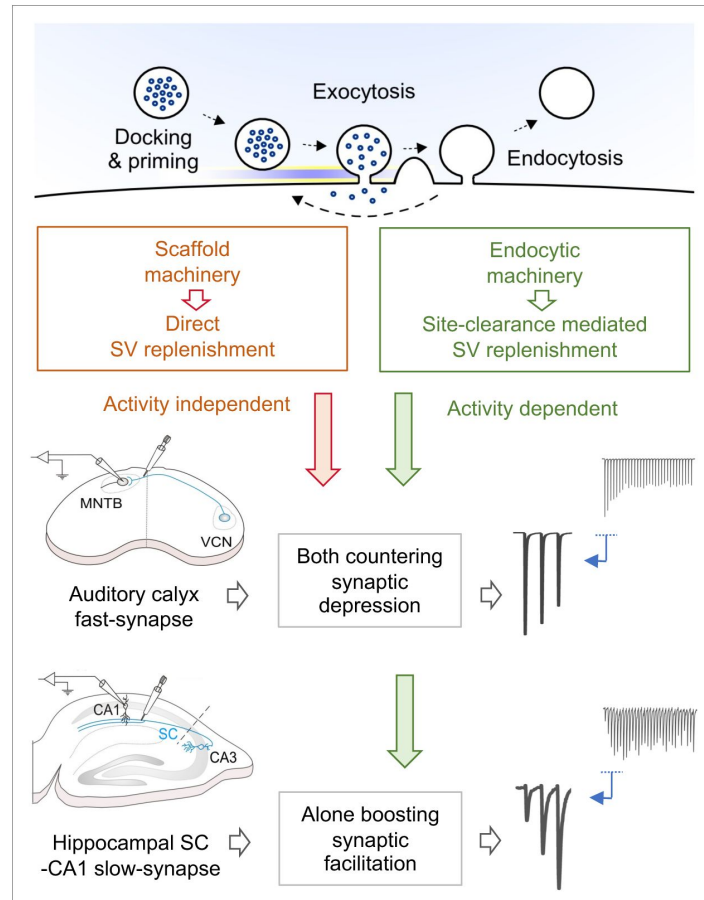


Figure 7.

Hypothetical vesicle replenishment scheme by endocytosis and scaffold-machineries during repetitive transmission at fast-signaling calyx and slow-plastic hippocampal CA1 synapses.

During high-frequency transmission, endocytosis driven site-clearance allows activity-dependent replenishment of new vesicles to release sites. This endocytic function counteracts synaptic depression caused by vesicle depletion, thereby maintain synaptic strength, enabling high-fidelity fast neurotransmission at sensory relay synapses, like at the calyx of Held. This endocytosis driven synaptic strengthening function augments synaptic facilitation at slow-plastic synapses like hippocampal CA1 synapses that exhibit long-term plasticity, thereby boosting its induction capability for memory formation. Whereas, the presynaptic scaffold machinery plays a powerful direct vesicle replenishment role, independent of endocytosis and activity, thereby rapidly translocating new vesicles to open release sites including those just opened by endocytic site-clearance. This scaffold function is specifically devoted to fast-signaling synapses, with high release probability but not to plastic synapses, where vesicle depletion is minimal due to low release probability.

Materials and Methods

Animals

All experiments were performed in accordance with the guidelines of the Physiological Society of Japan and animal experiment regulations at Okinawa Institute of Science and Technology Graduate University. C57BL/6J mice of either sex after hearing onset (postnatal day 13-15) were used for the experiment and animals were maintained on a 12-hour light/dark cycle with food and water *ad libitum*.

Method Details

Slice preparation (brainstem and hippocampal) and electrophysiology

Following decapitation of C57BL/6J mice under isoflurane anesthesia, brains were isolated and transverse slices (200 μm thick) containing the medial nucleus of the trapezoid body (MNTB) or parasagittal slices (300 μm) containing the hippocampus were cut using a vibratome (VT1200S, Leica) in ice-cold artificial cerebrospinal fluid (aCSF, see below) with reduced Ca^{2+} (0.1 mM) and increased Mg^{2+} (2.9 mM) concentrations and with NaCl replaced by 200 mM sucrose. Slices were incubated for 1 h at 37°C in standard aCSF containing (in mM): 115 NaCl, 2.5 KCl, 25 NaHCO_3 , 1.25 NaH_2PO_4 , 1.3 or 2 CaCl_2 , 1 MgCl_2 , 10 glucose, 3 myo-inositol, 2 sodium pyruvate, and 0.4 sodium ascorbate (pH 7.3-7.4 when bubbled with 95% O_2 and 5% CO_2 , 285-290 mOsm). Unless otherwise noted, EPSCs were recorded in 1.3 mM $[\text{Ca}^{2+}]$ aCSF at 37°C. During recording, a brain slice in a temperature-controlled chamber (RC-26GLP, PM-1; Warner instruments) was continuously perfused with aCSF solutions without (control) or with the pharmacological blockers using a peristaltic pump (5-6 $\mu\text{l/s}$). Recordings with or without the blockers were made from separate cells in different slices from a minimum of 3-4 animals for each set of experiment. In experiments at PT, solutions kept in a water bath (37°C) and passed through an in-line heater (SH-27B, Warner instruments) placed immediately before the recording chamber and maintained at 37°C ($\pm 0.2^\circ\text{C}$) by a controller (TC-344C, Warner instruments). For RT experiments, no heating was used, and the temperature was within 22-24°C. Recordings in the presence of pharmacological blockers were made within 10-60 min of drug application and for control experiments a slice was kept in the recording chamber no longer than 1 h.

Whole-cell recordings were made using a patch-clamp amplifier (EPC-10 USB, HEKA Elektronik, Germany) from the calyx of Held presynaptic terminals, postsynaptic MNTB principal neurons, or hippocampal CA1 pyramidal neurons visually identified with a 60X water immersion objective (LUMPlanFL, Olympus) attached to an upright microscope (BX51WI, Olympus, Japan). Data were acquired at a sampling rate of 50 kHz using Patchmaster software (for EPC-10 USB) after online filtering at 5 kHz. The patch pipettes were pulled using a vertical puller (PIP6, HEKA) and had a series resistance of 3-4 M Ω .

Membrane capacitance recordings from presynaptic calyx terminal (at PT)

The patch pipettes for presynaptic capacitance recording were wax coated to reduce stray capacitance and had a series resistance of 6-15 M Ω , which was compensated by up to 60% for its final value to be 6 M Ω . To isolate presynaptic Ca^{2+} currents (I_{Ca}) during membrane capacitance measurements (C_m), tetrodotoxin (1 μM) and tetra-ethyl-ammonium chloride (TEA, 5 mM) were routinely added to block Na^+ and K^+ channels, respectively. Calyx terminals were voltage clamped at a holding potential of -70 mV and a sine wave (peak to peak = 60 mV at 1 kHz) was applied. A single pulse (5 ms) or a train of 20 ms pulses (15 stimuli at 1 Hz or 10 stimuli at 10 Hz) to +10 mV were given to induce slow, fast-accelerating, and fast endocytosis. The capacitance trace within 450 ms after the stimulation pulse was excluded from analysis to avoid capacitance artifacts (Yamashita et al., 2005 [43](#)). The exocytic capacitance jump (ΔC_m) in response to a stimulation pulse

was measured as difference between baseline and mean C_m at 450-460 ms after the pulse onset. The endocytic rate was measured from the C_m decay phase 0.45-5.45 s from the end of 5 ms stimulation pulse, 0.45-0.95 s for 1 Hz train stimulation and 0.45-1.45 s after the last (10th) pulse for 10 Hz train stimulation. C_m traces were averaged at every 10 ms (for 10 Hz train stimulation) or 20 ms (for 5 ms single or 1 Hz train stimulation).

EPSC recordings from brainstem and hippocampus

For EPSC recordings, brainstem MNTB principal neurons or hippocampal CA1 pyramidal cells were voltage-clamped at the holding potential of -80 mV. To evoke EPSCs a bipolar electrode was placed on afferent fibers between MNTB and midline in brainstem slices or on the Schaffer collateral fibers in the stratum radiatum near the CA1 border in hippocampal slices.

The patch pipettes had a series resistance of 5-15 M Ω (less than 10 M Ω in most cells) that was compensated up to 80% for a final value of ~3 M Ω for EPSC recording from MNTB neurons. For EPSC recording from hippocampal pyramidal cells, no compensation was made, and capacitance artifacts were manually corrected. Stimulation pulses were applied through an isolator (Model 2100, A-M Systems) controlled by EPC10 amplifier (HEKA).

During EPSC recordings, strychnine-HCl (Sigma, 2 μ M), bicuculline-methiodide (Sigma, 10 μ M) and D-(-)-2-Amino-5-phosphonopentanoic acid (D-AP5, Tocris, 50 μ M) were added to isolate AMPA receptor-mediated EPSCs. In some experiments, to minimize AMPA receptor saturation and desensitization, kynurenic acid (Sigma, 1 mM) was added to aCSF. Unless otherwise specified, the internal pipette solution for presynaptic capacitance measurements contained (in mM): 125 Cs gluconate, 10 HEPES, 20 TEA-Cl, 5 sodium phosphocreatine, 4 Mg-ATP, 0.3 Na-GTP and 0.5 EGTA (pH 7.2-7.3, adjusted with CsOH, adjusted to 305-315 mOsm by varying Cs-gluconate concentration). For EPSC recording, the internal pipette solution contained QX-314 bromide (2 mM) and EGTA, either 5 mM (for MNTB cells) or 0.5 mM (for pyramidal cells), but otherwise identical to the presynaptic pipette solution. Dynasore (100 μ M), Pitstop-2 (25 μ M), ML 141 (10 μ M), Folimycin (also known as Concanamycin A; 67 nM), Latrunculin-A (20 μ M), Latrunculin B (15 μ M), and Bafilomycin A1 (5 μ M) were dissolved in DMSO (final concentration 0.1%, except for Latrunculin-A, 0.2%). Since Latrunculin A, Latrunculin-B and Folimycin are light sensitive, precautions were taken to minimize light exposure during handling and recording. Dynamin 1 proline-rich domain (PRD) peptide (sequence: PQVPSRPNRAP; GenScript) was dissolved in presynaptic pipette solution (1 mM). Membrane capacitance (C_m) recordings in the presence of Dyn-1 PRD peptide were initiated 4-5 min after whole-cell membrane rupture to allow its time of diffusion inside presynaptic terminals.

Data Analysis and Statistics

Experiments were designed as population study using different cells from separate brain slices under control and drug treatment, rather than on a same cell before and after the drug exposure. All data were analyzed using IGOR Pro (WaveMetrics) and Microsoft Excel. All values are given as mean $\pm$ SEM and significance of difference was evaluated by Student's unpaired *t*-test, unless otherwise noted. Significance level was set at $p < 0.05$, denoted with asterisks (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Data and Resources Availability

- All data generated during this study are included in the manuscript.
- Further information about resources should be addressed to authors.

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Reviewer #1 (Public Review):

Summary:

The study examines the role of release site clearance in synaptic transmission during repetitive activity under physiological conditions in two types of central synapses, calyx of Held and hippocampal CA1 synapses. After acute block of endocytosis by pharmacology, deeper synaptic depression or less facilitation was observed in two types of synapses. Acute block of CDC42 and actin polymerization, which possibly inhibits the activity of Intersectin, affected synaptic depression at the calyx synapse, but not at CA1 synapses. The data suggest an unexpected, fast role of the site clearance in counteracting synaptic depression.

Strengths:

The study uses acute block of the molecular targets with pharmacology together with precise electrophysiology. The experimental results are clear cut and convincing. The study also examines the physiological roles of the site clearance using action potential-evoked transmission at physiological Ca and physiological temperature at mature animals. This condition has not been examined.

Weaknesses:

Pharmacology may have some off-target effects, though acute manipulation should be appreciated and the authors have tried several reagents to verify the overall conclusions.

<https://doi.org/10.7554/eLife.90497.2.sa2>

Reviewer #2 (Public Review):

Summary:

In this manuscript, Mahapatra and Takahashi report on the physiological consequences of pharmacologically blocking either clathrin and dynamin function during compensatory endocytosis or of the cortical actin scaffold both in the calyx of Held synapse and hippocampal boutons in acute slice preparations

Strengths:

Although many aspects of these pharmacological interventions have been studied in detail during the past decades, this is a nice comprehensive and comparative study, which reveals some interesting differences between a fast synapse (Calyx of Held) tuned to reliably transmit at several 100 Hz and a more slow hippocampal CA1 synapse. In particular the authors find that acute disturbance of the synaptic actin network leads to a marked frequency-dependent enhancement of synaptic depression in the Calyx, but not in the hippocampal synapse. This striking difference between both preparations is the most interesting and novel finding.

Weaknesses:

Unfortunately, however, these findings concerning the different consequences of actin depolymerization are not sufficiently discussed in comparison to the literature. My only criticism concerns the interpretation of the ML 141 and Lat B data. With respect to the Calyx data, I am missing a detailed discussion of the effects observed here in light of the different RRP subpools SRP and FRP. This is very important since Lee et al. (2012, PNAS 109 (13) E765-E774) showed earlier that disruption of actin inhibits the rapid transition of SRP SVs to the FRP at the AZ. The whole literature on this important concept is missing. Likewise, the role of actin for the replacement pool at a cerebellar synapse (Miki et al., 2016) is only mentioned in half a sentence. There is quite some evidence that actin is important both at the AZ (SRP to FRP transition, activation of replacement pool) and at the peri-active zone for compensatory endocytosis and release site clearance. Both possible underlying mechanisms (SRP to FRP transition or release site clearance) should be better dissected.

<https://doi.org/10.7554/eLife.90497.2.sa1>

Reviewer #3 (Public Review):

The manuscript by Mahapatra and Takahashi addresses the role of presynaptic release site clearance during sustained synaptic activity. The authors characterize the effects of pharmacologically interfering with SV endocytosis (pre-incubation with Dynasore or Pitstop-2) on synaptic short-term plasticity (STP) at two different CNS synapses (calyx of Held synapses and hippocampal SC to CA1 synapses) using patch-clamp recordings in acute slices under experimental conditions designed to closely mimic a physiological situation (37°C and 1.3 mM external $[Ca^{2+}]$). Endocytosis blocker-induced changes in STP and in the recovery from short-term depression (STD) are compared to those seen after pharmacologically inhibiting actin filament assembly (pre-incubation with Latrunculin-B or the selective Cdc42 GTPase inhibitor ML-141). Presynaptic capacitance (C_m) recordings in calyx terminals were used to establish the effects of the pharmacological maneuvers on SV endocytosis.

Latrunculin-B and ML-141 neither affect SV endocytosis (assayed by C_m recordings) nor EPSC recovery following conditioning trains, but strongly enhances STD at calyx synapses. No changes in STP were observed at Latrunculin-B- or ML-141-treated SC to CA1 synapses.

Dynasore and Pitstop-2 slow down endocytosis, limit the total amount of exocytosis in response to long stimuli, enhance STD in response to 100 Hz stimulation, but profoundly accelerate EPSC recovery following conditioning 100 Hz trains at calyx synapses. At SC to CA1

synapses, Dynasore and Pitstop-2 reduce the extent of facilitation and lower relative steady-state EPSCs suggesting a change in the facilitation-depression balance in favor of the latter.

The authors use state-of-the-art techniques and their data, which is clearly presented, leads to authors to conclude that endocytosis is universally important for clearance of release sites while the importance of scaffold protein-mediated site clearance is limited to 'fast synapses'.

Unfortunately, and perhaps not completely unexpected in view of the pharmacological tools chosen, there are several observations which remain difficult to understand:

(1) Blocking site clearance affects release sites that have previously been used, i.e. sites at which SV fusion has occurred and which therefore need to be cleared. Calyces use at most 20% of all release sites during a single AP, likely fewer at 1.3 mM external $[Ca^{2+}]$. Even if all those 20% of release sites become completely unavailable due to a block of release site clearance, the 2nd EPSC in a train should not be reduced by >20% because ~80% of the sites cannot be affected. However, ~50% EPSC reduction was observed (Fig. 2B1, lower right panel) raising the possibility that Dynasore does more than specifically interfering with SVs endocytosis (and possibly Pitstop as well). Non-specific effects are also suggested by the observed two-fold increase in initial EPSC size in SC to CA1 synapses after Dynasore pre-incubation.

(2) More severe depression was observed at calyx synapses after blocking endocytosis which the authors attribute to a presynaptic mechanism affecting pool replenishment. When probing EPSC recovery after conditioning 100 Hz trains, a speed up was observed mediated by an "unknown mechanism" which is "masked in 2 mM $[Ca^{2+}]$ ". These two observations, deeper synaptic depression during 100 Hz but faster recovery from depression following 100 Hz, are difficult to align and no attempt was made to find an explanation.

(3) To reconcile previous data reporting a block of Ca^{2+} -dependent recovery (CDR) by Dynasore or Latrunculin (measured at 2 mM external $[Ca^{2+}]$) with the data presented here (using 1.3 mM external $[Ca^{2+}]$) reporting no effect or a speed up of recovery from depression, the authors postulate that "CDR may operate only when excessive Ca^{2+} enters during massive presynaptic activation" (page 10 line 244). While that is possible, such explanation ignores plenty of calyx studies demonstrating fiber stimulation-induced CDR and elucidating molecular pathways mediating fiber stimulation-induced CDR, and it also completely dismisses the strong change in recovery time course after 10 Hz conditioning (single exponential) as compared to 100 Hz conditioning (double exponential with a pronounced fast component).

Strong presynaptic stimuli such as those illustrated in Figs. 1B,C induce massive exocytosis. The illustrated C_m increase of 2 to 2.5 pF represents fusion of 25,000 to 30,000 SVs (assuming a single SV capacitance of 80 aF) corresponding to a 12 to 15% increase in whole terminal membrane surface (assuming a mean terminal capacitance of ~16 pF). Capacitance measurements can only be considered reliable in the absence of marked changes in series and membrane conductance. Documentation of the corresponding conductance traces is therefore advisable for such massive C_m jumps and merely mentioning that the first 450 ms after stimulation were skipped during analysis or referring to previous publications showing conductance traces is insufficient.

All bar graphs in Figures 1 through 6 and Figures S3 through S6 compare three or even four (Fig. 5C) conditions, i.e. one control and at least two treatment data sets. It appears as if repeated t-tests were used to run multiple two-group comparisons (i.e. using the same control data twice for two different comparisons). Either a proper multiple comparison test should be used or a Bonferroni correction or similar multiple-comparison correction needs to be applied.

Finally, the terminology of contrasting "fast-signaling" (calyx synapses) and "slow-plastic" (SC synapses) synapses seems to imply that calyx synapses lack plasticity, as does the wording "conventional bouton-type synapses involved in synaptic plasticity" (page 11, line 251). I assume, the authors primarily refer to the maximum frequencies these two synapse types typically transmit (fast-signaling vs slow-signaling)?

<https://doi.org/10.7554/eLife.90497.2.sa0>

Author Response

The following is the authors' response to the original reviews.

eLife assessment:

This important study combines a comparative approach in different synapses with experiments that show how synaptic vesicle endocytosis in nerve terminals regulates short-term plasticity. The data presented support the conclusions and make a convincing case for fast endocytosis as necessary for rapid vesicle recruitment to active zones. Some aspects of the description of the data and analysis are however incomplete and would benefit from a more rigorous approach. With more discussion of methods and analysis, this paper would be of great interest to neurobiologists and biophysicists working on synaptic vesicle recycling and short-term plasticity mechanisms.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The study examines the role of release site clearance in synaptic transmission during repetitive activity under physiological conditions in two types of central synapses, calyx of Held and hippocampal CA1 synapses. After the acute block of endocytosis by pharmacology, deeper synaptic depression or less facilitation was observed in two types of synapses. Acute block of CDC42 and actin polymerization, which possibly inhibits the activity of Intersectin, affected synaptic depression at the calyx synapse, but not at CA1 synapses. The data suggest an unexpected, fast role of the site clearance in counteracting synaptic depression.

Strengths:

The study uses an acute block of the molecular targets with pharmacology together with precise electrophysiology. The experimental results are clear-cut and convincing. The study also examines the physiological roles of the site clearance using action potential-evoked transmission at physiological Ca and physiological temperature at mature animals. This condition has not been examined.

Weaknesses:

Pharmacology may have some off-target effects, though acute manipulation should be appreciated. Although this is a hard question and difficult to address experimentally, reagents may affect synaptic vesicle mobilization to the release sites directly in addition to blocking endocytosis.

To acutely block vesicle endocytosis, we utilized two different pharmacological tools, Dynasore and Pitstop-2, after testing their blocking spectra and potencies at the calyx presynaptic terminals and collected data of their common effects on target functions. Since

the recovery from STD was faster at the calyx synapses in the presence of both endocytic blockers in physiological 1.3 mM [Ca²⁺] (Figure 2B), but not in 2.0 mM [Ca²⁺] (Figure S4), they might facilitate vesicle mobilization in physiological condition.

Reviewer #2 (Public Review):

Summary:

In this manuscript, Mahapatra and Takahashi report on the physiological consequences of pharmacologically blocking either clathrin and dynamin function during compensatory endocytosis or of the cortical actin scaffold both in the calyx of Held synapse and hippocampal boutons in acute slice preparations

Strengths:

Although many aspects of these pharmacological interventions have been studied in detail during the past decades, this is a nice comprehensive and comparative study, which reveals some interesting differences between a fast synapse (Calyx of Held) tuned to reliably transmit at several 100 Hz and a more slow hippocampal CA1 synapse. In particular, the authors find that acute disturbance of the synaptic actin network leads to a marked frequency-dependent enhancement of synaptic depression in the Calyx, but not in the hippocampal synapse. This striking difference between both preparations is the most interesting and novel finding.

Weaknesses:

Unfortunately, however, these findings concerning the different consequences of actin depolymerization are not sufficiently discussed in comparison to the literature. My only criticism concerns the interpretation of the ML 141 and Lat B data. With respect to the Calyx data, I am missing a detailed discussion of the effects observed here in light of the different RRP subpools SRP and FRP. This is very important since Lee et al. (2012, PNAS 109 (13) E765-E774) showed earlier that disruption of actin inhibits the rapid transition of SRP SVs to the FRP at the AZ. The whole literature on this important concept is missing. Likewise, the role of actin for the replacement pool at a cerebellar synapse (Miki et al., 2016) is only mentioned in half a sentence. There is quite some evidence that actin is important both at the AZ (SRP to FRP transition, activation of replacement pool) and at the peri-active zone for compensatory endocytosis and release site clearance. Both possible underlying mechanisms (SRP to FRP transition or release site clearance) should be better dissected.

The concept of FRP and SRP are derived from voltage-clamp step-depolarization experiments at calyces of Held in pre-hearing rodents at RT, which cannot be directly dissected in data of action-potential evoked EPSCs at post-hearing calyces at physiological conditions. However, we dissected as much by referring to related literatures in new paragraphs in Result section (p9-10), particularly on the different effects of Latrunculin application and experimental conditions by adding a new supplementary Figure (now S5). Regarding F-actin role in vesicle replenishment at cerebellar synapses, we added sentences in Discussion section (p14, last paragraph).

Reviewer #3 (Public Review):

General comments:

(1) While Dynasore and Pitstop-2 may impede release site clearance due to an arrest of membrane retrieval, neither Latrunculin-B nor ML-141 specifically acts on AZ scaffold proteins. Interference with actin polymerization may have a number of consequences many of which may be unrelated to release site clearance. Therefore, neither Latrunculin-

B nor ML-141 can be considered suitable tools for specifically identifying the role of AZ scaffold proteins (i.e. ELKS family proteins, Piccolo, Bassoon, α -liprin, Unc13, RIM, RBP, etc) in release site clearance which was defined as one of the principal aims of this study.

In this study, we focused our analysis on the downstream activity of scaffold protein intersectin by comparing the common inhibitory effects of CDC42 and actin polymerization, by use of ML141 and Latrunculin B, respectively, on vesicle endocytosis and synaptic depression/ facilitation without addressing diverse individual drug effects. To avoid confusion we removed “AZ” from scaffold protein.

(2) Initial EPSC amplitudes more than doubled in the presence of Dynasore at hippocampal SC->CA1 synapses (Figure S2). This unexpected result raises doubts about the specificity of Dynasore as a tool to selectively block SV endocytosis.

It is possible that Dynasore might have unknown or off-target effects. However, the main conclusion is backed up by Pitstop-2.

(3) In this study, the application of Dynasore and Pitstop-2 strongly decreases 100 Hz steady-state release at calyx synapses while - quite unexpectedly - strongly accelerates recovery from depression. A previous study found that genetic ablation of dynamin-1 actually enhanced 300 Hz steady-state release while only little affecting recovery from depression (Mahapatra et al., 2016). A similar scenario holds for the Latrunculin-B effects: In this study, Latrunculin-B strongly increased steady-state depression while in Babu et al. (2020), Latrunculin-B did not affect steady-state depression. In Mahapatra et al. (2016), Latrunculin-B marginally enhanced steady-state depression. The authors need to make a serious attempt to explain all these seemingly contradicting results.

The latrunculin effect on STD can vary according to the condition of application and external $[Ca^{2+}]$, which we show in a new supplemental Figure S5. The latrunculin effect on the recovery from STD also varies with temperature, $[Ca^{2+}]$, and animal age, which affect Ca^{2+} -dependent fast recovery component from depression. We added paragraphs for this issue in Results section (p9-10).

(4) The experimental conditions need to be better specified. It is not clear which recordings were obtained in 1.3 mM and which (if any?) in 2 mM external Ca. It is also unclear whether 'pooled data' are presented (obtained from control recordings and from separate recordings after pre-incubation with the respective drugs), or whether the data actually represent 'before'/'after' comparisons obtained from the same synapses after washing in the respective drugs. The exact protocol of drug application (duration of application/pre-incubation?, measurements after wash-out or in the continuous presence of the drugs?) needs to be clearly described in the methods and needs to be briefly mentioned in Results and/or Figure legends.

We added methodological explanations and reworded sentences in the text to be clear for pharmacological data derived from non-sequential separate experiments.

(5) The authors compare results obtained in calyx with those obtained in SC->CA1 synapses which they considered examples for 'fast' and 'slow' synapses, respectively. There is little information given to help readers understand why these two synapse types were chosen, what the attributes 'fast' and 'slow' refer to, and how that may matter for the questions studied here. I assume the authors refer to the maximum frequency these two synapse types are able to transmit rather than to EPSC kinetics?

Yes, the “fast and slow” naming features maximum operating frequency these synapses can transmit. We reworded “fast and slow” to “fast-signaling and slow-plastic” and added explanation in the text.

(6) Strong presynaptic stimuli such as those illustrated in Figures 1B and C induce massive exocytosis. The illustrated C_m increase of 2 to 2.5 pF represents a fusion of 25,000 to 30,000 SVs (assuming a single SV capacitance of 80 aF) corresponding to a 12 to 15% increase in whole terminal membrane surface (assuming a mean terminal capacitance of ~16 pF). Capacitance measurements can only be considered reliable in the absence of marked changes in series and membrane conductance. Since the data shown in Figs. 1 and 3 are central to the argumentation, illustration of the corresponding conductance traces is mandatory. Merely mentioning that the first 450 ms after stimulation were skipped during analysis is insufficient.

Conductance trace is shown with a trace of capacitance change induced by a square pulse in our previous paper (Yamashita et al, 2005 Science).

(7) It is essential for this study to preclude a contamination of the results with postsynaptic effects (AMPA saturation and desensitization). AMPAR saturation limits the amplitudes of initial responses in EPSC trains and hastens the recovery from depression due to a 'ceiling effect'. AMPAR desensitization occludes paired-pulse facilitation and reduces steady-state responses during EPSC trains while accelerating the initial recovery from depression. The use of, for example, 1 mM kynurenic acid in the bath is a well-established strategy to attenuate postsynaptic effects at calyx synapses. All calyx EPSC recordings should have been performed under such conditions. Otherwise, recovery time courses and STP parameters are likely contaminated by postsynaptic effects. Since the effects of AMPAR saturation on EPSC₁ and desensitization on EPSC_{ss} may partially cancel each other, an unchanged relative STD in the presence of kynurenic acid is not necessarily a reliable indicator for the absence of postsynaptic effects. The use of kynurenic acid in the bath would have had the beneficial side effect of massively improving voltage-clamp conditions. For the typical values given in this MS (10 nA EPSC, 3 MOhm Rs) the expected voltage escape is ~30 mV corresponding to a change in driving force of 30 mV/80 mV=38%, i.e. initial EPSCs in trains are likely underestimated by 38%. Such large voltage escape usually results in unclamped $I_{Na(V)}$ which was suppressed in this study by routinely including 2 mM QX-314 in the pipette solution. That approach does, however, not reduce the voltage escape.

Glutamate released during AP-evoked EPSCs does not saturate or desensitize postsynaptic receptors at post-hearing calyces of Held (Ishikawa et al, 2002; Yamashita et al, 2003) although it does in pre-hearing calyces (Yamashita et al, 2009). In fact, as shown in Figure S3, our results are essentially the same with or without kynurenate.

(8) In the Results section (pages 7 and 8), the authors analyze the time course into STD during 100 Hz trains in the absence and presence of drugs. In the presence of drugs, an additional fast component is observed which is absent from control recordings. Based on this observation, the authors conclude that '... the mechanisms operate predominantly at the beginning of synaptic depression'. However, the consequences of blocking or slowing site clearing are expected to be strongly release-dependent. Assuming a probability of <20% that a fusion event occurs at a given release site, >80% of the sites cannot be affected at the arrival of the second AP even by a total arrest of site clearance simply because no fusion has yet occurred. That number decreases during a train according to $(1-0.2)^n$, where n is the number of the AP, such that after 10 APs, ~90% of the sites have been used and may potentially be unavailable for new rounds of release after slowing

site clearance. Perhaps, the faster time course into STD in the presence of the drugs isn't related to site clearance?

Enhanced depression at the beginning of stimulation indicates the block of rapid SV replenishment mechanism, which includes endocytosis-dependent site-clearance and scaffold-dependent vesicle translocation to release sites.

(9) In the Discussion (page 10), the authors present a calculation that is supposed to explain the reduced size of the second calyx EPSC in a 100 Hz train in the presence of Dynasore or Pitstop-2. Does this calculation assume that all endocytosed SVs are immediately available for release within 10 ms? Please elaborate.

We do not assume rapid endocytosed vesicle reuse within 10 ms as it requires much longer time for glutamate refilling (7s at PT; Hori & Takahashi, 2012). Instead, already filled reserved vesicles can rapidly replenish release sites if sites are clean and scaffold works properly. Results shown in Figure S6 also indicate that block of vesicle transmitter refilling has no immediate effect on synaptic responses.

(10) It is not clear, why the bafilomycin/folimylin data is presented in Fig. S5. The data is also not mentioned in the Discussion. Either explain the purpose of these experiments or remove the data.

These v-ATPase blockers, which block vesicular transmitter refilling, are reported to enhance EPSC depression at hippocampal synapses at RT and 2 mM [Ca²⁺] presumably because of lack of filled vesicles undergoing rapid vesicle recycling (eg Kiss & Run). We thought it important to determine whether these data have physiological relevance since such a mechanism might also regulate synaptic strength during repetitive transmission. However, our results did not support its physiological relevance. Since these results are not within our main questions, the negative results are shown in supplementary Figure 6 and explained in the last paragraph of Result section (p11), but were not discussed further in Discussion section.

(11) The scheme in Figure 7 is not very helpful.

We updated the scheme to summarize our conclusion that vesicle replenishment through endocytosis-dependent site-clearance and scaffold-dependent mechanism independently co-operate to strengthen synaptic efficacy during repetitive transmission at calyx fast-signaling synapses. However, endocytic site clearance is solely required to support facilitation at slow-plastic hippocampal SC-CA1 synapses.

Recommendations for the authors:

First, my deep apologies for the long delay in reviewing your paper. All reviewers are now in agreement that the paper has valuable new information, but some methods are not described well and some results appear to be incompatible with previous results in the literature. The discussion of previous literature is also incomplete and not well-balanced. With more discussion of methods and literature strengthened this paper would be of great interest to neurobiologists and biophysicists working on synaptic vesicle recycling and short-term plasticity mechanisms. We ask that you address the comments and revise your paper before we can fully recommend the paper as being an important contribution with compelling evidence and a strong data set that supports the conclusions.

We explained methods more explicitly. Apparent incompatibility with previous results is now explained and discussed with new supplementary data.

(1) In this study, the application of Dynasore and Pitstop-2 strongly decreased 100 Hz steady-state release at calyx synapses while - quite unexpectedly - it strongly accelerated recovery from depression. A previous study found that genetic ablation of dynamin-1 actually enhanced 300 Hz steady-state release while only little affecting recovery from depression (Mahapatra et al., 2016). A similar scenario holds for the Latrunculin-B effects: In this study, Latrunculin-B strongly increased steady-state depression while in Babu et al. (2020), Latrunculin-B did not affect steady-state depression. In Mahapatra et al. (2016), Latrunculin-B marginally enhanced steady-state depression. The authors need to make a serious attempt to explain all these seemingly contradicting results.

Lack of change in the recovery from depression in dynamin-1 knockout mice by Mahapatra et al (2016) is consistent with results in Figure S4 in 2 mM [Ca²⁺], whereas accelerated recovery by Dynasore (Figure 2B2) is observed in 1.3 mM [Ca²⁺] suggesting that it is masked in 2 mM [Ca²⁺] but revealed in physiological [Ca²⁺] (p7, top paragraph). In both cases, however, recovery from STD is not prolonged unlike Hosoi et al (2009).

The latrunculin issues are discussed in Results section with newly added Supplementary Figure S5 (p9-10).

(2) The experimental conditions need to be better specified. It is not clear which recordings were obtained in 1.3 mM and which (if any?) in 2 mM external Ca. It is also unclear whether 'pooled data' are presented (obtained from control recordings and from separate recordings after pre-incubation with the respective drugs), or whether the data actually represent 'before'/'after' comparisons obtained from the same synapses after washing in the respective drugs. The exact protocol of drug application (duration of application/pre-incubation?, measurements after wash-out or in the continuous presence of the drugs?) needs to be clearly described in the methods and needs to be briefly mentioned in Results and/or Figure legends.

We made these points clearer in Method section and Result section.

(3) Please cite and discuss briefly previous papers that have shown fast endocytosis in the calyx of Held with membrane capacitance measurements like Renden and von Gersdorff, J Neurophysiology, 98:3349, 2007 and Taschenberger et al., Neuron, 2002. These papers first showed exocytosis and endocytosis kinetics in more mature (hearing) mice calyx of Held and at higher physiological temperatures.

One of these literatures relevant to the present study is quoted in p4.

(4) The findings concerning the different consequences of actin depolymerization are not sufficiently discussed in comparison to the literature. My only criticism concerns the interpretation of the ML 141 and Lat B data. With respect to the Calyx data, I am missing a detailed discussion of the effects observed here in light of the different RRP subpools SRP and FRP. This is very important since Lee et al. (2012, PNAS 109 (13) E765-E774) showed earlier that disruption of actin inhibits the rapid transition of SRP SVs to the FRP at the AZ. The whole literature on this important concept is missing. Likewise, the role of actin for the replacement pool at a cerebellar synapse (Miki et al., 2016) is only mentioned in half a sentence. There is quite some evidence that actin is important both at the AZ (SRP to FRP transition, activation of replacement pool) and at the peri-active zone for compensatory endocytosis and release site clearance. Both possible underlying mechanisms (SRP to FRP transition or release site clearance) should be better dissected.

We added discussions on the issue of latrunculin in Result section by quoting previous literatures (p9-10). Since there is no direct evidence (by vesicle imaging) for the presence of

FRP and SRP, these definitions derived from voltage clamp step-depolarization studies are difficult to incorporate into the dissection of synaptic depression in physiological conditions.

Reviewer #1 (Recommendations For The Authors):

I have no major comments, but the following issues may be addressed.

(1) The term "fast and slow" synapses may be relative and a bit confusing. I do not think hippocampal synapses are slow synapses.

We have replaced “fast and slow” by “fast-signaling and slow-plastic” to represent their functions and added explanation in the text.

(2) Off-target effects of pharmacological effects may be discussed. In this respect, bafilomycin experiments can be used to argue against the slow effects of vesicle cycling such as endocytosis, and vesicle mobilization. However, the effects on rapid vesicle mobilization cannot be excluded entirely. Because I cannot exclude the absence of off-target effects either (can be addressed by looking at single vesicle imaging at nano-scale, which is hard to do or looking at EM level quantitatively?), I feel this is a matter of discussion.

It is possible that Dynasore might have unknown or off-target effects. However, the main conclusion is backed up by Pitstop-2.

(3) Fig2 A2, B2 and Fig 4 A2 and B2. It is easier to plot the recovery only normalized to the initial value. Subtracting steady-state is somewhat confusing because the recovery looks faster after deeper depression, but this may be just apparent.

We have given values for both types of plots in Table 2, which indicates no essential difference in the recovery parameters.

Reviewer #2 (Recommendations For The Authors):

Line 51: Rajappa et al. (2016) investigated clearance deficits in synaptophysin KO mice (not synaptobrevin).

Corrected.

Line 54: intersectin is introduced as AZ scaffold protein, although in most of the literature, it is referred to as an endocytic scaffold protein (also in the cited one, e.g. Sakaba et al. 2013). At least, this should be discussed.

Since blockers of intersectin downstream protein activity has no effect on vesicle endocytosis (Figure 3 and Sakaba et al, 2013), we called it (presynaptic) scaffold protein instead of endocytic scaffold protein.

Reviewer #3 (Recommendations For The Authors):

Minor comments

Page 1, Title: I don't think the presented data address the role of the presynaptic scaffold in SV replenishment. In addition, 'SV replenishment' and 'site clearance' should not be used synonymously as it seems to be implied here.

In this study our focus was on the downstream activity of scaffold protein intersectin and since block of its downstream effector proteins CDC42 and actin activities do not obstruct the

endocytic activity (Fig 3, and Sakaba et al., 2013), instead of naming it as “endocytic scaffold protein”, we adopted “presynaptic scaffold protein”.

We have corrected it in the text.

Page 2, Abstract: Clarify 'physiologically optimized condition' here and elsewhere in the manuscript.

Abstract: in physiologically optimized condition → in physiological temperature and Ca²⁺.

Page 3, line 62: I don't think 'the site-clearance hypothesis is widely accepted'. There are very few models that implement such a mechanism. Examples would be Pan & Zucker (2009) Neuron and Lin, Taschenberger & Neher 2022 (PNAS) which could be cited.

62: the site-clearance hypothesis is “widely accepted” → “well supported”

Page 3 line 77: Please clarify 'fast synapses

77: fast synapses → fast-signaling synapses, added clarification in the text.

Page 4, line 100: Please clarify 'in the maximal rate'.

100: in the maxima rate → reached during 1-Hz stimulation.

Page 6, line 136: Please clarify 'to reduce the gap'.

136: To reduce the gap between these different results → To explore the reason for these different results

Page 7, line 157: I don't consider ML141 and Latrunculin-B 'scaffold protein inhibitors'.

157: scaffold protein inhibitors had no effect on → reworded as “none of these inhibitors affected fast or slow endocytosis”.

Page 7, line 162: P-value missing.

162: $p < 0.001$ added.

Page 8, line 184: "Since both endocytic blockers and scaffold inhibitors enhanced synaptic depression with a similar time course" consider rephrasing. Sounds like you refer to the time course by which these drugs exert their effect after being applied.

184: Since both endocytic blockers and scaffold inhibitors enhance synaptic depression with a similar time course → Since the enhancement of synaptic depression by endocytic blockers or scaffold inhibitor occurred mostly at the early phase of synaptic depression.

Same on page 11, line 250: "At the calyx of Held, scaffold protein inhibitors significantly enhanced synaptic depression with a time course closely matching to that enhanced by endocytic blocker" Please consider rephrasing.

At the calyx of Held, scaffold protein inhibitors significantly enhanced synaptic depression with a time course closely matching to that enhanced by endocytic blocker → the early phase of synaptic depression like endocytic blockers

Page 13, line 318: Please clearly state which experiments were performed at 1.3 mM and which at 2 mM external Ca if two different concentrations were used during recordings.

320: Added text “Unless otherwise noted, EPSCs were recorded in 1.3 mM [Ca²⁺] aCSF at 37°C” in the methods.

Page 15: line 346: Reference in the wrong format.

346; (25) → (Yamashita et al, 2005)

Page 15: line 351: Do you mean to say every 10 s and every 20 s? Please clarify.

No, averaged at 10 ms and 20 ms, respectively as written.

Page 16, line 369: 1 mM kyn was present in only very few experiments shown in the supplemental figures. Please clarify.

368: In some experiments, to test in the presence of 1 mM kyn, if there is any difference in enhanced STD following endocytic block. However, as shown in Figure S3, our results are essentially the same with or without kynurenate, suggesting glutamate released during AP-evoked EPSCs does not saturate or desensitize postsynaptic receptors at post-hearing calyces of Held (Ishikawa et al, 2002; Yamashita et al, 2003) unlike in pre-hearing calyces (Yamashita et al, 2009).

Page 16, line 387: You cannot simply use multiple t-tests to compare a single control to multiple test conditions which seems to be the scenario here. Please correct or clarify.

Experimental protocols are clarified in Methods as “Experiments were designed as population study using different cells from separate brain slices under control and drug treatment, rather than on a same cell before and after the drug exposure.”

Table S1: 'Endo decay rate'. It's either the 'Endo rate' or the 'Decay rate of delta Cm'. Please correct.

Corrected as Endocytosis rate (Endo rate).